

E- Portfolio Activity: Jaccard Distance/Dissimilarity Calculations

To complete the Jaccard Distance/Dissimilarity Calculations, I started by creating groups with only positive attributes for each name.

Jack (Fever, Test-1)

Mary (Fever, Test-1, Test-3)

Jim (Fever, Cough)

The two formulas that I need to take into consideration are:

Jaccard formula = $J(A, B) = \frac{|A \cap B|}{|A \cup B|}$

Distance formula = $D(A, B) = 1 - J(A, B)$

Moving on to calculating the Jaccard distance/dissimilarity, the calculations and the results are the following:

- Jack, Mary
 - They have two elements in common: fever and test-1
 - The union consists of three elements: fever, Test-1, and Test-3
 - The similarity is $2/3$
 - The distance calculation are as follows: $1 - 2/3 = 1/3 = 0.33$
- Jack, Jim
 - They have one element in common: Fever
 - The union consists of three elements: Fever, Test-1, and Cough
 - The similarity is $1/3$
 - The distance calculation are as follows: $1 - 1/3 = 2/3 = 0.67$

- Jim, Mary
 - They have one element in common: Fever
 - The union consists of four elements: Fever, Cough, Test-1, and Test-3
 - The similarity is $\frac{1}{4}$
 - The distance calculation are as follows: $1 - \frac{1}{4} = \frac{3}{4} = 0.75$

In conclusion, the Jaccard distance shows different levels of similarity between the pairs of individuals.

Jack and Mary have the highest similarity because they share two positive attributes, resulting in the lowest distance. In contrast, Jim and Mary have the highest distance, indicating that they are the least similar pair because they share only one common attribute. Overall, the Jaccard distance is useful for comparing pairs of observations based on the presence of common attributes.